

Passive Traffic Intelligence

Feature Guide and Working Capability Overview

SIH26145 · Unidirectional Data-Diode Cybersecurity Pipeline

OPERATING PRINCIPLE

Production traffic is observed through a one-way TAP or data diode. SentinelAI OS transforms passive packet and flow metadata into evidence-backed security intelligence, without probing, decrypting, blocking, or transmitting into the production network.

Working prototype · Streaming telemetry · Dashboard alerts · Metadata-only analysis

1. Solution overview

SentinelAI OS provides a monitoring-enclave pipeline for critical infrastructure gateways and peering links. It accepts packet captures, live passive interface observations, and normalized NetFlow/IPFIX/sFlow-style records; incrementally extracts metadata features; applies statistical, heuristic, and local anomaly scoring; then streams structured alerts to the SOC dashboard.

Passive security boundary

- Read-only ingest: data is received from a capture source, TAP, or one-way diode with no production return path.
- No active interaction: no probes, handshakes, packet injection, containment commands, or source/destination callbacks.
- No payload decryption: TLS and QUIC-oriented analysis is based solely on observable metadata and fingerprint fields.
- Forensic-ready workflow: flow and alert metadata are persisted with timestamps, identifiers, confidence, severity, and evidence.

Architecture

Stage	Working capability
1. Passive observation	PCAP upload/parser, Scapy capture on an approved ingest interface, and normalized flow-record API.
2. Streaming pipeline	Bounded queues and background workers process observations incrementally and broadcast updates through WebSockets.
3. Feature extraction	Rates, timing, entropy, DNS naming traits, port/host fan-out, byte asymmetry, SYN rate, and encrypted-session metadata.
4. Detection & scoring	Threat heuristics are fused with statistical and optional local Isolation Forest anomaly scoring.
5. SOC intelligence	Persisted structured alerts and a live dashboard show severity, confidence, evidence, flows, and pipeline state.

2. Threat detection capabilities

The prototype delivers the following detection categories through the same passive processing path used by simulated, replayed, uploaded-PCAP, and flow-record inputs.

Threat class	Passive indicators used	Intelligence output
Volumetric / protocol DDoS	Packets-per-second, bytes-per-second, SYN rate, protocol profile, high-rate flow characteristics.	SYN flood, UDP amplification, and spoofed-source-flood alerts with measured rates.
Botnet C2 beaconing	Inter-arrival mean and standard deviation; recurring low-variance flow timing.	C2 beacon alert with periodicity evidence and confidence score.
DGA and DNS tunnelling	DNS-label entropy, uncommon character n-grams, digit ratio, query length, and query frequency.	DGA or DNS tunnel alert showing anomalous domain/query features.
Encrypted-session malware	JA3/JA3S/JA4-ready fields, TLS version/cipher fields, packet count, timing regularity, and port 443 metadata.	Encrypted-session anomaly alert; no plaintext inspection or decryption.
Reconnaissance / scanning	Unique destination ports, unique destination hosts, flow fan-out, and scan velocity.	Reconnaissance alert with affected-port/host evidence.
Data exfiltration	Protected-network direction, outbound/inbound byte counts, byte ratio, and high-volume flow behavior.	Data-exfiltration alert with byte-volume and asymmetry evidence.

All detections are passive observations. Recommended actions are investigation and evidence preservation—not inline intervention.

3. Models, features, and scoring

SentinelAI OS uses a practical ensemble approach suitable for an offline monitoring enclave. Deterministic rules provide immediate coverage while statistical and ML-assisted scoring establish additional anomaly context.

Feature engineering

- Flow volume: packet count, byte count, duration, packets/second, and bytes/second.
- Timing: mean and standard deviation of inter-arrival time, enabling beacon regularity analysis.
- Network behavior: distinct ports and destinations, protocol, TCP SYN rate, and direction.
- DNS behavior: query labels, entropy, n-gram rarity, query length, frequency, and record-type metadata.
- Encrypted metadata: JA3, JA3S, JA4, TLS version, cipher fields, packet timing, and packet-size/rate context.
- Exfiltration behavior: outbound and inbound bytes plus outbound-to-inbound ratio.

Inference and confidence

Rules emit per-threat scores from observed evidence. The fusion stage selects the highest-scoring applicable detection, maps it to LOW/MEDIUM/HIGH/CRITICAL severity, and produces a confidence score. A local Isolation Forest can be fitted after benign observations are collected, providing an additional metadata-only anomaly score without a cloud dependency.

Alert schema

Required field	Provided by SentinelAI OS
Timestamp	UTC event timestamp
Flow identifier	Stable generated or supplied flow ID
Threat class	Normalized detection label
Confidence score	0–1 fused confidence plus risk score/severity
Supporting evidence	Detection features, rates, entropy, timing, byte ratios, and model/detection method

4. Dashboard and operational workflow

The Unidirectional Defense dashboard presents the live passive-traffic operating picture for analysts.

- Data-diode status panel: read-only state, inbound-only direction, return-path blocked, zero active probes, and decryption disabled.
- Live flow stream: source/destination, protocol, byte count, packet rate, and timestamped metadata observations.
- Threat investigation queue: threat class, severity, confidence percentage, source/destination, detection method, and supporting evidence.
- PCAP analysis: upload a capture and create metadata-only flow and alert results without replaying traffic.
- Live passive ingest: select an allow-listed TAP/data-diode receive interface and start bounded passive capture.
- Replay and simulation: demonstrate normal traffic and each attack category through the real detector path.
- WebSocket updates: dashboard metrics, flows, and alerts update in near real time.

Input and deployment options

Input	Use
PCAP/PCAPNG/CAP	Offline forensic or replay analysis; only metadata is retained for detection.
Live capture NIC	Passive receive-side interface for a TAP or data diode. Interface allow-listing is supported.
Normalized flow API	Integration point for NetFlow, IPFIX, sFlow, or an enclave-side collector/diode relay.
Synthetic/replay feed	Safe functional demonstrations, analyst training, and controlled scenario validation.

5. Performance, validation, and deployment

The pipeline is designed for incremental processing rather than end-of-run reporting. The runtime exposes a local benchmark operation that sends a chosen number of flows through the real passive analysis path and records elapsed time and measured flows per second.

Throughput demonstration

Metric	Working implementation
Throughput unit	Measured flows per second, with packets/second and bytes/second also visible on the dashboard.
Benchmark path	On-demand benchmark endpoint exercises feature extraction, inference, persistence, and alert output locally.
Latency design	Bounded ingest queue plus background worker processing and live WebSocket delivery.
Operational protection	Bounded capture and analysis queues report depth and preserve capture continuity if downstream analysis is degraded.

Validation approach

Functional validation uses synthetic scenarios for normal traffic, SYN flood, UDP amplification, spoofed-source flood, C2, DGA, DNS tunnelling, encrypted-session anomaly, port/host scan, and exfiltration. The repository also contains focused unit coverage for the required demo categories. In a deployment, local benchmark results should be recorded for the actual enclave hardware and traffic profile.

Deployment checklist

- Place SentinelAI OS in the monitoring enclave only; connect its capture NIC to the receive side of the TAP/data diode.
- Configure UNIDIRECTIONAL_ALLOWED_INTERFACES with approved receive-only interfaces.
- Configure UNIDIRECTIONAL_PROTECTED_CIDRS to enable direction-aware exfiltration analysis.
- Ensure there is no route, management tunnel, packet injection capability, or mitigation integration back into production.
- Use PCAP/replay and the benchmark workflow to demonstrate expected detection behavior and measured flows-per-second capacity.

Conclusion

SentinelAI OS delivers a working passive cybersecurity intelligence prototype for one-directional critical-infrastructure monitoring. It ingests observable traffic metadata, detects and scores the required threat classes in near real time, and provides structured, evidence-backed analyst alerts through a live dashboard—while maintaining the read-only, no-decryption, no-return-path security boundary.